Time Series Analysis and Forecasting

A student-friendly introduction to time patterns, forecasting models, and implementation

1. What is a time series?

A **time series** is a set of measurements recorded over time, usually at regular intervals. Examples include daily temperature, monthly product sales, hourly electricity use, weekly website visits, and yearly population. **Time series analysis** looks for patterns in these measurements. **Forecasting** uses past patterns to estimate future values. A forecast is an estimate, not a guarantee.

2. Where is forecasting used?

Area	Example forecast
Business	Sales, inventory, customer demand
Weather	Temperature and rainfall
Energy	Electricity demand and production
Transport	Traffic and passenger numbers
Healthcare	Patient visits and disease trends
Finance	Prices and economic indicators

3. Components of a time series

A series may contain a long-term direction, regular repeating patterns, longer cycles, and irregular changes. A useful introductory model is:

Observed value = Trend + Seasonality + Cycle + Random variation

This additive form is useful when seasonal changes stay about the same size. If seasonal changes grow as the overall values grow, a multiplicative form may fit better.

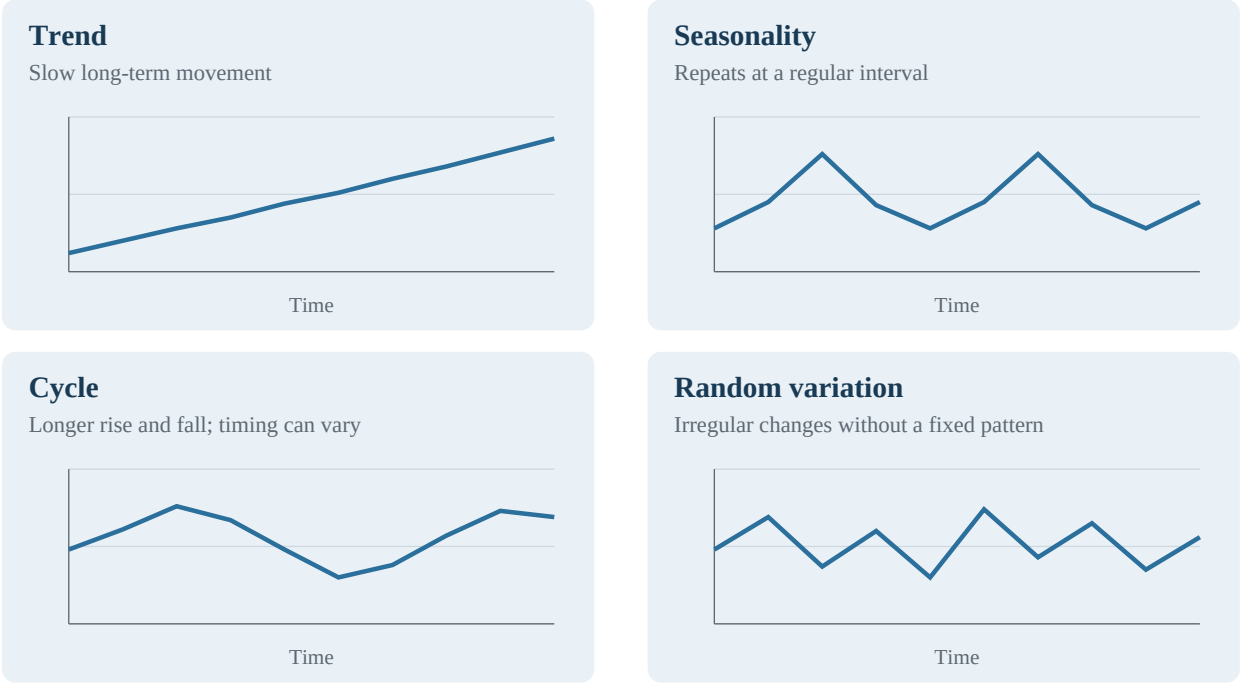

Figure 1. Illustrative patterns for the four components. These are teaching sketches, not data from a real study.

Trend: the long-term direction, such as shop sales gradually increasing over several years.

Seasonality: a pattern that repeats at a known interval, such as higher sales each December. For monthly data, one yearly seasonal period is 12 months.

Cycle: a longer rise and fall whose timing is not fixed, such as changes during economic expansions and slowdowns. **Random variation:** irregular movement left after the main patterns, such as an unusual drop in visits caused by a storm.

4. Forecasting models

ARIMA and exponential smoothing are two common model families. They make different assumptions, so it is good practice to compare candidate models on held-out data.

ARIMA

ARIMA means **AutoRegressive Integrated Moving Average**. It is written as **ARIMA(p, d, q)**: p is the number of past values used by the AR part, d is the number of times the data is differenced, and q is the number of past forecast errors used by the MA part.

- **AR - AutoRegressive:** uses earlier observed values to help predict the current value. For example, this month's sales may be related to last month's sales. AR(1) uses one previous time step.
- **I - Integrated:** means differencing the data to make its level more stable before modelling. First difference = current value minus previous value. If sales rise from 120 to 135, the difference is 15. d=1 means difference once.
- **MA - Moving Average:** uses earlier forecast errors to help predict the current value. If a model repeatedly underestimates sales after a promotion, information from past errors can affect its next estimate. Here, MA refers to past errors, not the simple moving-average smoothing calculation.

Example: ARIMA(1, 1, 1) for monthly sales uses one past value, differences the data once, and uses one past forecast error. These are example settings; the best settings depend on the data.

Exponential smoothing

Exponential smoothing gives more weight to recent observations. The weights on older observations decrease over time. Simple Exponential Smoothing is for data without a clear trend or seasonality; Holt's method includes trend; Holt-Winters includes trend and seasonality. For example, monthly sales that rise over time and also peak every December may be modelled with Holt-Winters.

5. Basic implementation in Python

A practical workflow is: put observations in time order; set the date index and frequency; plot the series; split earlier observations for training and later observations for testing; fit candidate models; forecast the test period; compare predictions with actual values; and inspect errors for remaining patterns. Never shuffle time-series observations for this evaluation.

```
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.holtwinters import ExponentialSmoothing

# sales: a chronological series of monthly sales
train = sales.iloc[:-12]
test = sales.iloc[-12:]  # hold out the final year

# ARIMA example
arima_fit = ARIMA(train, order=(1, 1, 1)).fit()
arima_forecast = arima_fit.forecast(steps=len(test))

# Holt-Winters with yearly monthly seasonality
hw_fit = ExponentialSmoothing(
    train, trend="add", seasonal="add", seasonal_periods=12
).fit()
hw_forecast = hw_fit.forecast(steps=len(test))
```

The settings above demonstrate how to call the models; they are not guaranteed to be best for every dataset. Compare against the test data with MAE (mean absolute error) or RMSE (root mean squared error). Choose using performance on data the model did not train on.

Quick recap

A time series records values over time. Its components include trend, seasonality, cycle, and random variation. ARIMA uses past values, differencing, and past errors; exponential smoothing emphasizes recent observations. Reliable evaluation keeps the test period later in time than the training period.

References for the Python API

statsmodels documentation: ARIMA model and Holt-Winters ExponentialSmoothing. See the official API pages at statsmodels.org/stable/generated/statsmodels.tsa.arima.model.ARIMA.html and statsmodels.org/stable/generated/statsmodels.tsa.holtwinters.ExponentialSmoothing.html.